Guneev Pannu

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Education

University of Toronto – 3.96 GPA Sept 2024 – May 2028

HBSc Double Major in Computer Science and Statistics, Math Minor

Relevant Courses: Data Structures & Algorithms, Operating Systems, Systems Programming, Databases, Security

Experience

Teaching Assistant Sept 2026 – Dec 2026

University of Toronto *Toronto, Canada*

- Will lead weekly tutorials for **40+** students, Python fundamentals (control flow, functions, lists) for CSC108
- Held **2 hours** of office hours per week, walking students through their code so they could debug issues themselves

Product Manager Sept 2026 – Present

UofT Machine Learning Club

- Leading club website's migration to **Payload CMS**, replacing Google Forms and Git-push content updates with a single self-serve platform for **11 execs** and **100+ applicants** per year
- Interviewed all club VPs across engineering, research, academics, etc, to gather requirements for the website redesign, and combined input into a prioritized roadmap for the year.

Web Developer May 2026 – Aug 2026

Professor - University of Toronto

- Built a live queuing and logging system using **Next.js**, **React**, **Docker**, and **PostgreSQL** to improve structure in office hours by reducing crowding, and by keeping attendance logs in case of cheating in **500+ student** class
- Worked in a team of 4, meeting weekly with supervising professor to discuss progress and decide next steps
- Wrote GitHub Action to automatically publish updates via self hosted runner to save time during development
- Automated student enrollment, saving **2+ hrs** of manual entry for professors by building a CSV pipeline
- Integrated UofT SAML authentication for **15K+ students**, eliminating all manual account creation

CS Research Assistant Sept 2026 – Dec 2026

University of Toronto *Toronto, Canada*

- Authored weekly lab materials and slide decks in LaTeX to deepen understanding of **500+ students** for CSC108

Projects

In Class Live QA Tool | *Next.js, Typescript, PostgreSQL, Redis, Docker*

- Architected a real-time UofT Q&A platform in **Next.js** securing a Fall rollout targeting **500+ students**
- Implemented quality of life features like chat notifications, and automatic question context with **React Hooks** to make it easier to monitor chat during lecture
- Streamed real time events using **WebSockets** to minimize delay in room events such as question posts, question upvotes, slide changes, and population count in **50+ student** rooms with average latency of **40ms** during load test

Blog Website Backend | *Spring Boot, Java, Maven, Git, PostgreSQL, Docker*

- Built a backend software application in a week that could support a blog post social media app, like Tumblr
- Integrated Spring Data JPA to simplify communication from the service layer to the persistence layer
- Implemented JWT-based auth using Spring Security so draft blogs can only be read and posted by their owner

Paint Program | *Java, JavaFX, Maven, Git*

- Built a paint program that allowed users to draw **180+ shapes** and lines on a resizable canvas using **JavaFX**
- Applied the **MVC** design pattern reducing merge conflicts by **30%**, allowing for more rapid development

Technical Skills

Languages: Typescript, Java, Python
Frameworks: Next.js, React, Node.js, Junit, Springboot, Agile
Developer Tools: Cursor, Claude Code, Git, Docker, PostgrsSQL, Postman, Linux
Libraries: pytest, Prisma, PyTorch, JavaFX